# TEAM DESCRIPTION PAPER

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## *Short Team Description*

We, Tux_3, compete in the RoboCupJunior Soccer Entry League and were formed in 2026 by merging two previously independent teams.

In the 2025 season, the members competed separately as Tux_3 (Alexander and Henning) and Tux_4 (Max and Toni), both using LEGO-based robots. At the Berlin qualification tournament, Tux_3 achieved 2nd place and Tux_4 achieved 4th place. Both teams qualified for the German Championship in Nuremberg, where Tux_3 finished 8th (one match away from European qualification) and Tux_4 finished 13th.

In 2026, the two teams combined their experience and formed a single team under the name Tux_3, redesigning the robot with fully custom 3D-printed components. At the Berlin qualification tournament 2026, the new team achieved 2nd place.

## *Short Robot Description*

The Tux_3 robot is a custom-built RoboCupJunior Entry League soccer robot designed for reliability, robustness, and continuous improvement. After competing with LEGO-based robots in 2025, the team fully redesigned the system for the 2026 season using custom 3D-printed components and a self-developed electronic architecture.

The robot is based on a two-motor differential drive system with integrated ball control and goal detection mechanisms. The overall design focuses on mechanical stability, precise ball handling, and structured decision-making during matches.

## *Hardware*

**Drive System**

- 2× Pololu 25Dx48L MP 12V motors
- Differential drive configuration
- Infineon IFX9201SG motor drivers

**Ball Handling**

- GA16Y-050-CE motor for dribbler
- Infineon BTN8982TA driver for kicker
- Light barrier for ball possession detection

**Sensors**

- Vishay TSSP6038TR infrared sensors for ball detection
- CMPS14 compass module for orientation stabilization
- Pixy2 camera for goal detection
- DFRobot_URM09 Ultrasonic sensors for distance measurement and wall correction

**Electronics**

- ESP32 microcontroller

- Custom wiring and structured component layout
- 3D-printed chassis designed in Autodesk Fusion

More about Hardware in '/Hardware/README.md'

## ***Software***

The software is structured around a high-frequency main control loop running on the ESP32. The system separates sensor acquisition, decision logic, and motor control into modular components. Our program is based on the CoRoSoN-Kit (https://github.com/ItzJannis/CoRoSoN-Package_Draft).

Key characteristics:

- State-based decision logic:

```
enum State {
  Setzen,      //Reset
  Drehen,      //Spin
  Angreifen,   //Attack
  Notfall,     //Emergency
  Befreien     //Rescue
};
```

- Ball tracking using infrared sensors
- PID Control:

```
float computePID(int error) {
  Iterm += error;
  Iterm = constrain(Iterm, -200, 200);

  float Dterm = error - lastDiff;
  lastDiff = error;

  return kP * error + kI * Iterm + kD * Dterm;
}
```

- Goal alignment using camera and compass
- Custom stuck-detection and recovery behavior:

```
void checkSteckenGeblieben() {                                          //Check if you are stuck
  if (millis() - steckenTimer >= 1750) {
    if (abs(winkelFehler(steckenKompass, compass.Angle())) <= 2) {
      befreienTimer = millis();
      steckenGeblieben = true;
    } else {
      steckenKompass = compass.Angle();
      steckenTimer = millis();
    }
  }
}
```

- Wall correction using ultrasonic feedback

The software was developed with a focus on maintainability, clarity, and match robustness.

More about the software in: [https://github.com/alexander0038/TDP-Robocup-Tux_3/tree/main/software](https://github.com/alexander0038/TDP-Robocup-Tux_3/tree/main/software)

## ***Challenges During the Season***

During the transition from LEGO to a fully custom 3D-printed robot, several challenges were encountered:

1. Mechanical Stability:
   Early prototypes suffered from structural weaknesses and misalignment in the dribbler system.
2. Different Programming Languages:
   We started with the programming language NXC, which translates to “Not exactly C”, and later switched to C/C++.
3. Sensor Calibration:
   Reliable ball detection and consistent compass readings required extensive testing and calibration.
4. Differential Drive Precision:
   Achieving accurate turning and goal alignment without omni wheels required significant tuning of motor control.

## ***Challenges Successfully Mastered***

- Structural redesign improved mechanical robustness.
- Stable ball control was achieved through optimized dribbler positioning.
- Sensor placement and filtering significantly increased detection reliability.
- Stuck-detection logic improved match recovery performance.

- Team collaboration improved after merging two former teams into one unified structure.

These improvements led to a 2nd place finish at the 2026 Berlin qualification tournament.

## ***Future Plans***

For the next season, the team plans to:

- Compete in Robocup Junior Soccer Lightweight.
- Learn more about the hardware.
- Deepen our knowledge about programming
- Improve our ability to analyze and find mistakes in hardware and software faster
- Improve teamwork even further (if 's even possible)

We hope that one day we will compete in the World Championship and win first place. What sounds like a dream today might come true one day.